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MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

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ASSIGNMENT: 1

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NepalRide Premium Car Rental System CET138 Full Stack Development

Student: 240704403 | Module Leader: David Grey | Submission: June 2026 What

is CSS.

Live demo link: https://rihanashrestha567-star.github.io/Demos/css_demo.html

CSS is the language used to describe the presentation and visual styling of HTML documents.

It controls colors, fonts, spacing, layout, animations and responsive behavior. CSS is used to separate content from design allowing the same HTML structure to be styled differently for devices and contexts.

Modern CSS3 has introduced some features including flexbox and grid for layouts, custom properties, keyframe animations,

transitions transforms and advanced selectors. These features enable developers to create interactive interfaces without relying heavily on JavaScript.

CSS also includes design capabilities through media queries allowing NepalRide websites to adapt gracefully across mobile phones, tablets and desktop screens.

Why CSS Matters in Modern Web Development

CSS determines how users perceive and interact with the NepalRide website.

Good CSS creates polished experiences that build trust and engagement.

Poor CSS leads to layouts, inconsistent styling and frustrated users.

Modern CSS techniques like custom properties and animations have made it possible to create app-like interfaces with minimal JavaScript.

CSS is also critical for performance.

Written CSS loads faster and renders more efficiently than JavaScript-based styling solutions.

How NepalRide Demonstrates CSS

NepalRide showcases CSS through its dark theme with amber and teal accents, glassmorphism effects using backdrop blur and extensive keyframe animations.

The hero section features a bouncing car with spinning wheels moving road stripes and parallax scrolling mountains - all created with pure CSS.

The impressive demonstration is the interactive CSS Playground in the NepalRide Portfolio section.

Users can control the cars speed, suspension bounce and neon underglow color using realtime sliders that update CSS variables and animation durations dynamically.

The NepalRide project uses CSS custom properties for theming, spacing and even physics values like bounce height making the entire design system easily adjustable from one central location.

Code Example 1. Core Implementation

```
:root { primary:
#0f172a;
--accent-amber: #f59e0b;
--bounce-height: 2px;
}

.hero-animated-car { animation: carVibe
0.15s infinite alternate; }

@keyframes carVibe {
```

```
0% { transform: translateY(0); }

100% { transform: translateY( calc(-1
* var(--bounce-height))); }

}
```

Code Example 2. Interactive Feature

```
/* Interactive CSS Playground. Live variable updates */

function updateCssPlayground() { const speed =
document.getElementById('slideSpeed').value;
document.documentElement.style.setProperty(
'--bounce-height' `${bounce}px`); document.getElementById('cssWheel1').style
.animationDuration = `${speed * 0.6}s`;
}
```

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Visual Demonstrations & Screenshots from NepalRide

The NepalRide CSS implementation is most visible in the hero section animations. The interactive CSS Playground.

The NepalRide Portfolios CSS demo pane allows users to manipulate car physics and colors in time providing clear evidence of deep CSS understanding.

Glassmorphism effects, smooth hover transitions on car cards and the responsive grid layouts demonstrate professional-grade CSS

skills across the NepalRide application.

Figure 1: CSS Interactive Playground

Real-time sliders control car speed, suspension bounce and neon underglow color.

JavaScript updates CSS variables and animation durations live.

Shows understanding of CSS physics and custom properties.

Figure 2: Hero Section Animations

Pure CSS keyframe animations create the bouncing car, spinning wheels moving road stripes and parallax mountains.

No JavaScript is used for these animations.

Key Features & Technical Implementation Details

NepalRide uses CSS techniques extensively: custom properties for theming and physics values keyframe animations for all

movement effects, flexbox and grid for responsive layouts, glassmorphism with backdropfilter and CSS transforms for smooth interactions.

The CSS Playground demonstrates manipulation of animation-duration and custom properties through JavaScript.

The NepalRide project also implements breakpoints for mobile, tablet and desktop with a mobile-first hamburger menu that appears below 768px screen width.

Implemented Features in NepalRide:

- CSS custom properties for theming and dynamic physics values Challenges Faced

& Solutions Applied

- Challenge: Making CSS animations feel natural and performant.

Solution: Used transform and opacity properties of changing layout properties and kept animations under 0.6s duration.

Challenge: Creating car movement with CSS only.

Solution: multiple animations (bounce + wheel spin + road movement) with different timings for a natural feel.

Challenge: Making the CSS Playground intuitive.

Solution: Used labeled sliders with live value displays and immediate visual feedback on the demo car.

Keyframe animations for car bounce, wheel spin, road movement parallax

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Personal Reflection & Learning Outcomes

CSS used to feel like the part of web development to me.

After building NepalRide I now see CSS as one of the creative and powerful tools available.

The ability to create physics simulations using only CSS variables and JavaScript updates was eye-opening.

The CSS Playground was the rewarding part to build.

Watching users control the cars speed and colors in time made me appreciate how powerful modern CSS has become.

I learned that good CSS is about more than making things pretty. Its about creating systems that are maintainable performant and delightful to use.

Learning Outcomes Achieved:

- Demonstrated CSS3 techniques including custom properties and keyframe animation

NepalRide showcases strong CSS skills through its extensive use of modern techniques, interactive playground and polished

visual design.

The CSS Playground, in particular provides evidence of deep understanding.

This foundation enables the creation of performant and engaging web interfaces.

NepalRide demonstrates how CSS can be used to create an user experience.

The NepalRide project is an example of how CSS can be used to create a professional and engaging web interface.

CSS is a powerful tool and NepalRide is a great example of how it can be used.

